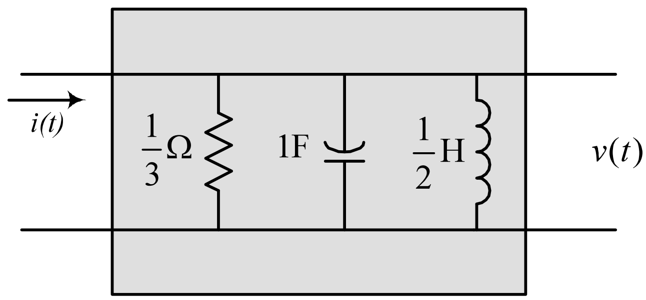
Laplace Transforms - Solution Sheet

1) A resistor, capacitor and inductor having values of $\frac{1}{3}$ Ω, 1F and $\frac{1}{2}$ H respectively, are connected in parallel. A unit ramp of current is applied at time $t = 0$ and there are zero initial conditions. Calculate the resulting voltage across the circuit (i) by solving the associated differential equation in full, and (ii) by starting with the Laplace version of the circuit.

Solution

(i) Differential equation approach...



We have (KCL) $$i = i_R + i_L + i_C = \frac{v}{R} + i_L + C\frac{dv}{dt}$$

Where i_R, i_C and i_L are the currents through the resistor, capacitor and inductor branches respectively. Since $v = L\frac{di_L}{dt}$, this gives a second order differential equation for the inductor current:

$$\frac{d^2 i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{1}{LC}i_L = \frac{1}{LC}i \quad \text{in general.}$$

In this particular case we have $$\frac{d^2 i_L}{dt^2} + 3\frac{di_L}{dt} + 2i_L = 2t \quad \text{for } t \geq 0.$$

The complementary equation is $s^2 + 3s + 2 = 0 \Longrightarrow (s+1)(s+2) = 0 \Longrightarrow s = -1, -2$

So that the complementary function is $i_{L_{CF}}(t) = A\exp(-t) + B\exp(-2t)$

where *A* and *B* are arbitrary constants which depend on the initial conditions.

Choose a trial particular integral of the form $i_{L_{PI}}(t) = Ct + D$

where *C* and *D* are undetermined coefficients. Substituting this into the original DE gives

$$0 + 3C + 2(Ct + D) = 2t \implies C = 1, D = -3/2$$

So that the general solution for the inductor current is

$$i_L(t) = A\exp(-t) + B\exp(-2t) + t - 3/2$$

We know that $i_L(0) = 0$ and that $\frac{di_L}{dt}(0) = 0$, so that the arbitrary constants must satisfy:

$$0 = A + B - \frac{3}{2} \text{ and } 0 = -A - 2B + 1 \Longrightarrow A = 2, B = -\frac{1}{2}$$

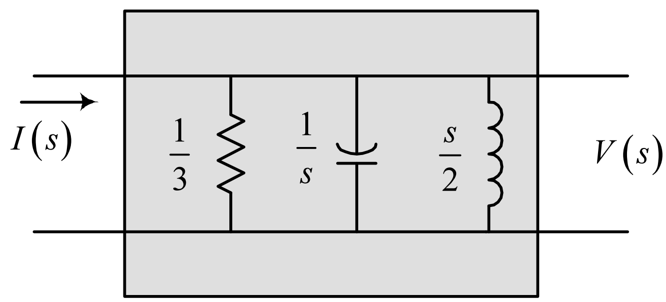
Hence the particular solution is $i_L(t) = 2\exp(-t) - \frac{1}{2}\exp(-2t) + t - 3/2$

We need the output voltage in fact, and since $v = \frac{1}{2}\frac{di_L}{dt}$ the required solution is

$$\boxed{v(t) = -\exp(-t) + \frac{1}{2}\exp(-2t) + \frac{1}{2} \qquad t \geq 0}$$

(ii) Laplace component approach...

The Laplace version of the circuit (with ZIC) is



The combined Laplace impedance of the three components is

$$\frac{1}{3 + s + \frac{2}{s}} = \frac{s}{s^2 + 3s + 2}$$

so that $$V(s) = \frac{s}{s^2 + 3s + 2} I(s) = \frac{s}{(s+1)(s+2)} I(s)$$

In this particular case

$$i(t) = t \quad \Longrightarrow \quad I(s) = \frac{1}{s^2}$$

So that the Laplace output is $V(s) = \frac{s}{(s+1)(s+2)}\frac{1}{s^2} = \frac{1}{s(s+1)(s+2)}$

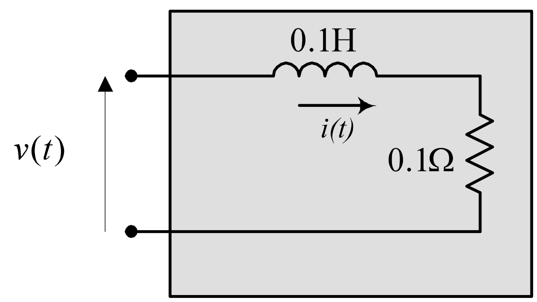
Using partial fractions $$V(s) = \frac{\frac{1}{2}}{s} + \frac{-1}{(s+1)} + \frac{\frac{1}{2}}{(s+2)}$$

so that inverting gives $$v(t) = -\exp(-t)u(t) + \frac{1}{2}\exp(-2t)u(t) + \frac{1}{2}u(t)$$

or $$v(t) = -\exp(-t) + \frac{1}{2}\exp(-2t) + \frac{1}{2} \quad \text{for } t \geq 0.$$

Note that the Laplace approach makes it clear that a pole-zero cancellation occurs - generally this will be imperfect and, in reality, there will be some residual amount of a ramp component in the solution.
Note also that the Laplace approach is much faster/easier.

2) A bolt of lightning striking an overhead power line may be modelled as shown below. If the lightning voltage is of the form $v(t) = t\exp(-t)$ and there are zero initial conditions, (i) solve the differential equation for the circuit to determine the current which flows, and (ii) repeat the calculation starting with the Laplace version of the circuit.



<u>Solution</u>

(i) Differential equation approach...

We have (KVL) $$v = iR + L\frac{di}{dt} \quad \text{in general.}$$

and in this specific case $$0.1\frac{di}{dt} + 0.1\,i = t\exp(-t) \quad \Rightarrow \frac{di}{dt} + i = 10\,t\exp(-t)$$

The complementary function/particular integral approach will work, but the quickest approach for this form of first-order differential equation is to multiply through by an integrating factor of $\exp(t)$.

$$\frac{di}{dt} + i = 10t\exp(-t) \quad \Rightarrow \frac{di}{dt}\exp(t) + i\exp(t) = 10t \quad \Rightarrow \frac{d}{dt}[i\exp(t)] = 10t$$

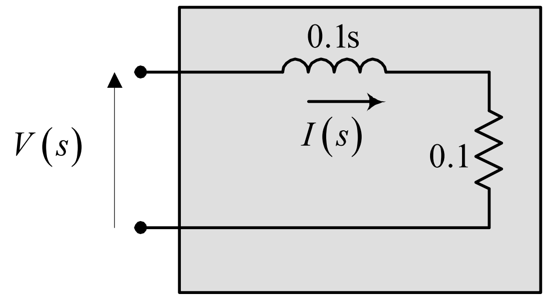
Integrating gives $$i\exp(t) = 5t^2 + A$$

and the fact that $i(0) = 0$ fixes the arbitrary constant of integration to be zero. Hence the output is

$$\boxed{i = 5t^2 \exp(-t) \qquad \text{for } t \geq 0.}$$

(ii) Laplace component approach...

The Laplace version of the circuit (with ZIC) is



The combined Laplace impedance of the two components is

$$0.1 + 0.1s$$

so that $$V(s) = (0.1 + 0.1s)\, I(s) \quad \Longrightarrow \quad I(s) = \frac{10}{(s+1)} V(s)$$

In this particular case $$v(t) = t\exp(-t) \quad \Longrightarrow \quad V(s) = \frac{1}{(s+1)^2}$$

So that the Laplace output is $$I(s) = \frac{10}{(s+1)} \frac{1}{(s+1)^2} = \frac{10}{(s+1)^3}$$

Recall that $L[e^{at} f(t) u(t)] = F(s-a)$, so that in this case the $(s+1)$ factor corresponds to multiplication by an $\exp(-t)$ function in the time domain:

$$i(t) = 10 \exp(-t)\, L^{-1}\left[\frac{1}{s^3}\right]$$

Also, since $\int_0^1 f(\tau) d\tau = \frac{1}{s} F(s)$

$$L^{-1}\left[\frac{1}{s^3}\right] = \int_0^1 L^{-1}\left[\frac{1}{s^2}\right] d\tau = \int_0^1 \tau d\tau = \frac{1}{2} t^2$$

Hence the output is	$i = 5t^2 \exp(-t) \qquad \text{for } t \geq 0.$

3) From first principles, establish the Laplace transform of the function $f(t) = t(t+1)$.

Solution

By definition,

$$L[t(t+1)] = \int_0^\infty t(t+1)e^{-st}dt \;=\int_0^\infty (t^2+t)e^{-st}dt \;=\int_0^\infty t^2e^{-st}dt \;+\int_0^\infty te^{-st}dt$$

Consider first $\int_0^\infty t\, e^{-st}dt$ and use integration by parts, with $u = t$ and $\dfrac{dv}{dt} = e^{-st}$:

$$\int_0^\infty te^{-st}dt = \left[t\left(-\frac{e^{-st}}{s}\right)\right]_0^\infty - \int_0^\infty 1\left(-\frac{e^{-st}}{s}\right)dt = 0 + \frac{1}{s}\int_0^\infty e^{-st}dt = \left[-\frac{1}{s^2}e^{-s}\right]_0^\infty = \frac{1}{s^2}$$

Next, consider $\int_0^\infty t^2e^{-st}dt$ and use integration by parts, with $u = t^2$ and $\dfrac{dv}{dt} = e^{-st}$:

$$\int_0^\infty t^2e^{-st}dt \;= \left[t^2\left(-\frac{e^{-s}}{s}\right)\right]_0^\infty - \int_0^\infty 2t\left(-\frac{e^{-s}}{s}\right)dt \;= 0 + \frac{2}{s}\int_0^\infty t\,e^{-s}\,dt = \frac{2}{s}\left(\frac{1}{s^2}\right) = \frac{2}{s^3}$$

Hence	$L[t(t+1)] = \frac{2}{s^3} + \frac{1}{s^2}$

4) Show that if $L[f(t)] = F(s)$ then $L[tf(t)] = -\frac{dF}{ds}(s)$.
Hence find $L[t\cos(2t)]$.

Solution

By definition $\quad L[f(t)] = F(s) = \int_0^\infty f(t)e^{-st}dt$

so that

$$\frac{dF}{ds}(s) = \frac{d}{ds}\left[\int_0^\infty f(t)e^{-st}dt\right]$$
$$= \int_0^\infty f(t)\frac{d}{ds}(e^{-st})dt \;= -\int_0^\infty tf(t)e^{-st}dt$$

but $\quad \int_0^\infty tf(t)e^{-st}dt \;= L[tf(t)]$

so comparing gives

$$L[tf(t)] = -\frac{dF}{ds}(s) \quad \text{QED}$$

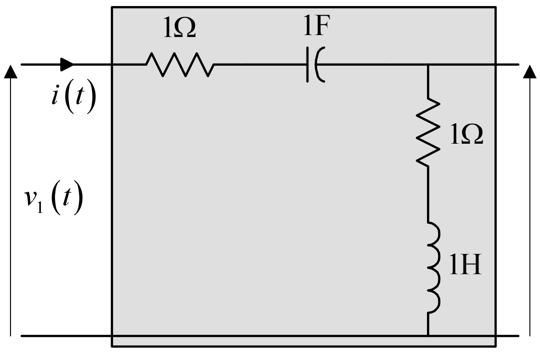
For the particular case here

$$L[t\cos(2t)] = -\frac{d}{ds}\left(L[\cos(2t)]\right)$$

$$= -\frac{d}{ds}\left(\frac{s}{s^2+4}\right) = -\frac{(s^2+4) - s(2s)}{(s^2+4)^2} = \frac{(s^2-4)}{(s^2+4)^2}$$

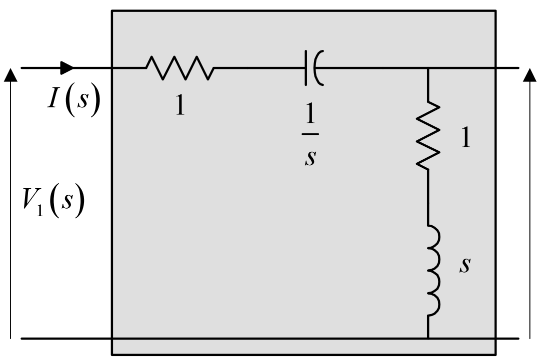
$$\boxed{L[t\cos(2t)\,] = \frac{(s^2-4)}{(s^2+4)^2}}$$

5) Evaluate $v_2(t)$ and $i(t)$ for the following circuit when a unit voltage step is applied as the input. Assume zero initial conditions.



Solution

The Laplace circuit is



Here, the Laplace impedance of the loop is $Z(s) = 1 + \frac{1}{s} + 1 + s = s + 2 + \frac{1}{s}$ where $V_1(s) = Z(s)I(s)$.

Hence $$I(s) = \frac{1}{Z(s)}V_1(s) = \frac{1}{s+2+\frac{1}{s}}V_1(s) = \frac{s}{s^2+2s+1}V_1(s) = \frac{s}{(s+1)^2}V_1(s)$$

Likewise $$V_2(s) = (s+1)I(s) = \frac{s(s+1)}{s^2+2s+1}V_1(s) = \frac{s}{(s+1)}V_1(s)$$

For the specific input $v_1(t) = u(t)$, $V_1(s) = \frac{1}{s}$, and so:

$$I(s) = \frac{s}{(s+1)^2}\frac{1}{s} = \frac{1}{(s+1)^2} \quad \Rightarrow \quad i(t) = te^{-t} \qquad \text{for } t \geq 0.$$

And

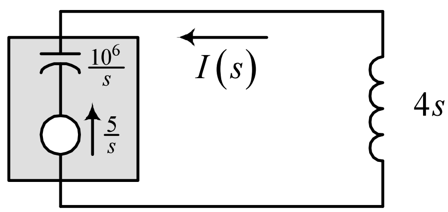
$$V_2(s) = \frac{s}{s+1}\frac{1}{s} = \frac{1}{s+1} \quad \Rightarrow \quad v_2(t) = e^{-t} \qquad \text{for } t \geq 0.$$

Note the two pole-zero cancellations - generally these will be imperfect due to issues such as component variations, and there will be some residual amounts of other modes in the solution.

6) A $1\mu\text{F}$ capacitor is charged to 5V, and is then connected across an inductor of value 4H. What current will subsequently flow?

Solution

In Laplace form the circuit is



The loop impedance is $\frac{10^6}{s} + 4s$ and the effective voltage source is $\frac{5}{s}$.

Hence

$$I(s) = \frac{-\frac{5}{s}}{\frac{10^6}{s} + 4s} = -\frac{5}{10^6 + 4s^2} = -\frac{\frac{5}{4}}{s^2 + 250 \times 10^3} = -\frac{\frac{5}{4}}{s^2 + (500)^2}$$

$$= -2.5 \times 10^{-3}\frac{500}{s^2 + (500)^2}$$

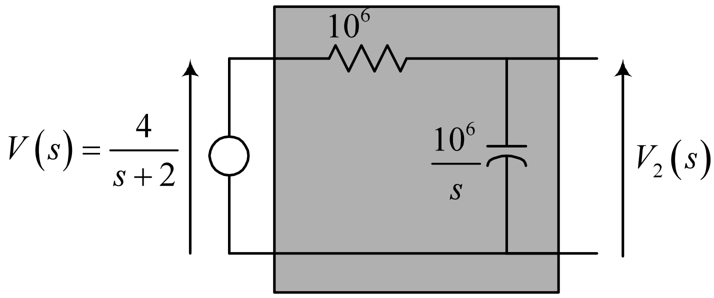
leading to

$$i(t) = -2.5 \times 10^{-3} \sin(500t) \text{ A} \quad \text{for } t \geq 0.$$

7) A voltage $v(t) = 4\exp(-2t)$, $(t \geq 0)$. is applied to a $1\text{M}\Omega$ resistor and a $1\mu\text{F}$ uncharged capacitor connected in series. Determine the resulting voltage across the capacitor.

Solution

In Laplace form the circuit is



Since $L[4\exp(-2t)\,u(t)] = \frac{4}{s+2}$.

The loop impedance is $Z(s) = 10^6 + \frac{10^6}{s}$ so that the loop current is

$$I(s) = \frac{1}{Z(s)}V(s) = \frac{1}{10^6 + \frac{10^6}{s}}\left(\frac{4}{s+2}\right) = \frac{4\times 10^{-6}s}{(s+1)(s+2)}$$

and so

$$V_2(s) = \frac{4\times 10^{-6}s}{(s+1)(s+2)}\left(\frac{10^6}{s}\right) = \frac{4}{(s+1)(s+2)}$$

$$= \frac{4}{(s+1)} - \frac{4}{(s+2)}$$

using partial fractions.

Hence, from tables: $v_2(t) = 4\exp(-t) - 4\exp(-2t)$ for $t \geq 0$.

8) Identify the zeroes and poles of the following functions:

(i) $\frac{s^2 + s}{s^2 + 5s + 6}$

(ii) $\frac{s^2 + 1}{s^3 - s}$

Solution

(i)

$$\frac{s^2 + s}{s^2 + 5s + 6} = \frac{s(s+1)}{(s+3)(s+2)}$$

So that the zeroes are $s = 0$ and $s = -1$ and the poles are $s = -2$ and $s = -3$.

(ii)

$$\frac{s^2+1}{s^3-s}=\frac{(s+j)(s-j)}{s(s+1)(s-1)}$$

So that the zeroes are $s=\pm j$ and the poles are $s=0$ and $s=\pm 1$.

9) Find the inverse Laplace transforms of the functions:

(i) $\dfrac{s+1}{s^2+5s+6}$

(ii) $\dfrac{s(s-1)}{(s+1)(s+2)(s+3)}$

(iii) $\dfrac{s-1}{(s+1)^2}$

(iv) $\dfrac{6s}{(s+1)^3}$

(v) $\dfrac{1}{s^3+s^2}$

Solution

(i)

$$\frac{s+1}{s^2+5s+6}=\frac{s+1}{(s+2)(s+3)}=\frac{-1}{(s+2)}+\frac{2}{(s+3)},\qquad \text{using partial fractions.}$$

Hence

$$L^{-1}\left[\frac{s+1}{s^2+5s+6}\right]=-\exp(-2t)+2\exp(-3t)\qquad t\geq 0$$

(ii)

$$\frac{s(s-1)}{(s+1)(s+2)(s+3)}=\frac{1}{(s+1)}+\frac{-6}{(s+2)}+\frac{6}{(s+3)},\qquad \text{using partial fractions.}$$

Hence

$$L^{-1}\left[\frac{s(s-1)}{(s+1)(s+2)(s+3)}\right]=\exp(-t)-6\exp(-2t)+6\exp(-3t)\qquad t\geq 0$$

(iii)

$$\frac{s-1}{(s+1)^2}=\frac{s+1-2}{(s+1)^2}=\frac{s+1}{(s+1)^2}+\frac{-2}{(s+1)^2}=\frac{1}{(s+1)}+\frac{-2}{(s+1)^2},\qquad \text{using partial fractions.}$$

Hence

$$L^{-1}\left[\frac{s-1}{(s+1)^2}\right] = \exp(-t) - 2t\exp(-t) \qquad t \geq 0.$$

(iv)

$$\frac{6s}{(s+1)^3} = \frac{0}{(s+1)} + \frac{6}{(s+1)^2} + \frac{-6}{(s+1)^3}, \qquad \text{using partial fractions.}$$

Hence

$$L^{-1}\left[\frac{6s}{(s+1)^3}\right] = 6t\exp(-t) - 3t^2\exp(-t) \qquad t \geq 0.$$

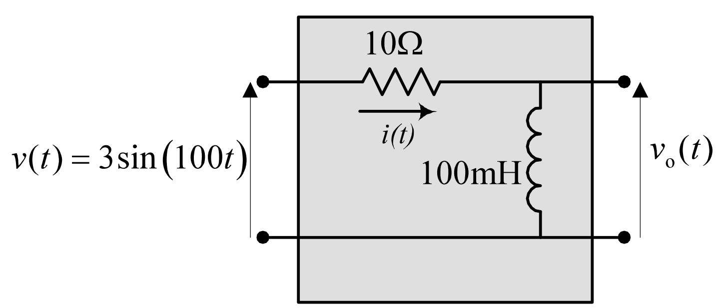
(iv)

$$\frac{1}{s^3+s^2} = \frac{1}{s^2(s+1)} = \frac{-1}{s} + \frac{1}{s^2} + \frac{1}{(s+1)}, \qquad \text{using partial fractions.}$$

Hence

$$L^{-1}\left[\frac{1}{s^3+s^2}\right] = -1 + \mathrm{t} + \exp(-t) \qquad t \geq 0.$$

10) Evaluate both the steady-state and transient responses for the following circuit:



<u>Solution</u>

The Laplace version of the circuit is